

EE4725 Quasi Optical Systems

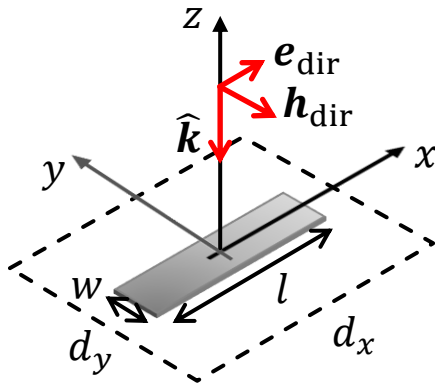
Matlab exercise: Scattering and Patterns from an FSS

Roderick G. Tapia Barroso

Problem 1

1. Calculate the frequency variation (6 to 14 GHz) of the reflection coefficient for the FSSs in the figures below, under plane-wave illumination.

Dipole strips in free space



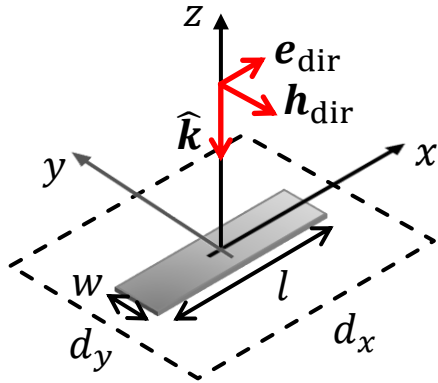
As an example, consider the following geometric parameters

```
dx = 15*mm;    theta = 0;  
dy = 15*mm;    phi   = 0;  
w  = 1 *mm;    TM incidence  
l  = 14*mm;
```

2. Calculate the variation of the reflection coefficient for the FSSs as a function of the angle of incidence at 10 GHz, in the planes $\phi = 0^\circ$ and $\phi = 90^\circ$. What is the maximum angle of incidence in the two planes for which the FSS still reflects more than 90%

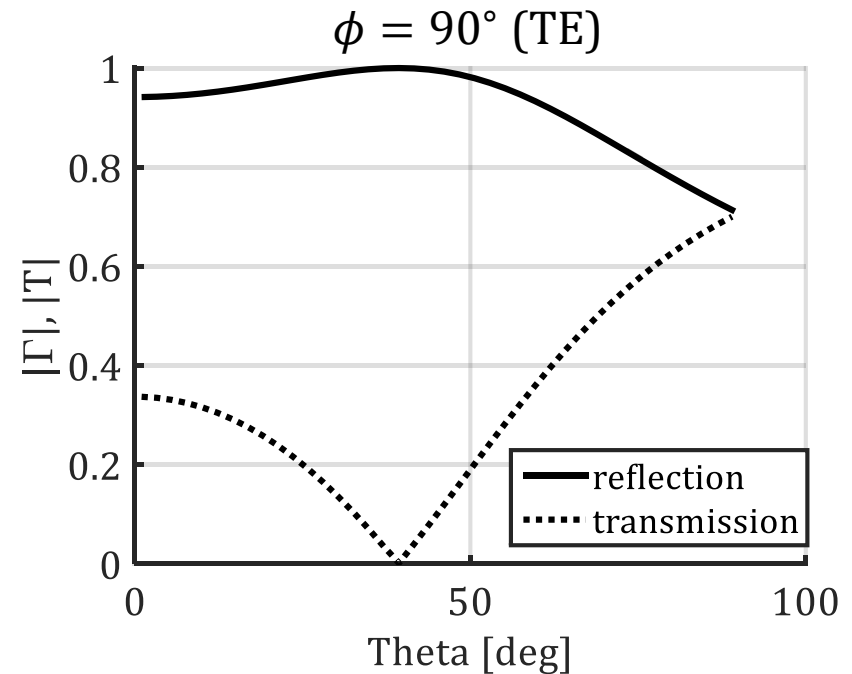
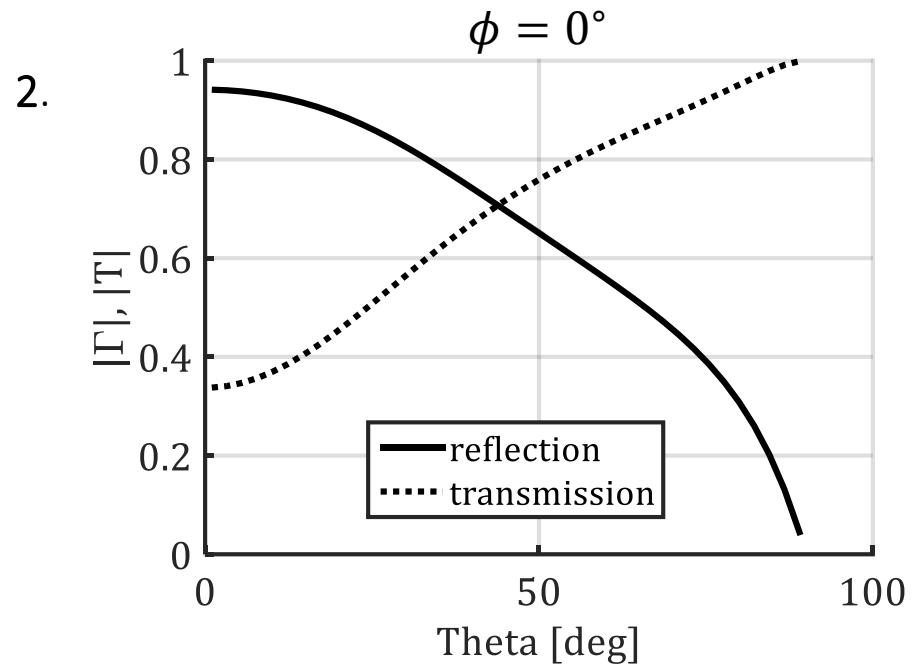
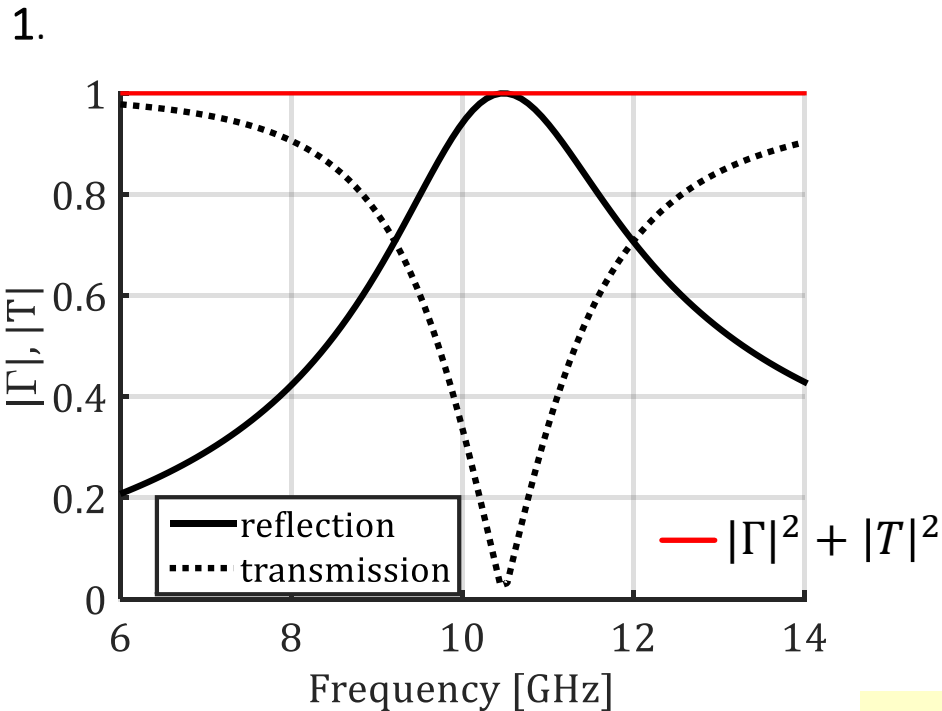
Problem 1: Solution

Dipole strips in free space



```
dx = 15*mm;
dy = 15*mm;
w = 1 *mm;
l = 14*mm;

theta = 0;
phi = 0;
TM incidence
```

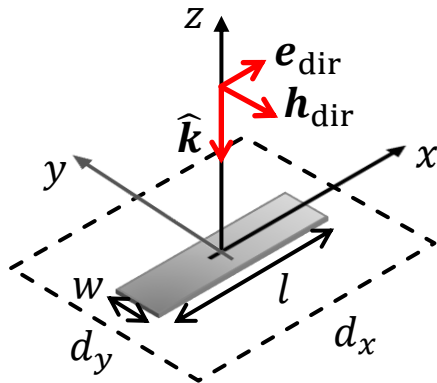


Why TE reflects more than TM at wide incidence angles?

Problem 2

Calculate the radiation patterns from the FSS in the figure below, under plane-wave illumination. Use a windowing technique with 10×10 unit cells.

Dipole strips in free space



As an example, consider the following geometric parameters

```
dx = 20*mm;    theta = 0;  
dy = 20*mm;    phi   = 0;  
w  = 1 *mm;    TM incidence  
l  = 15*mm;  
Freq = 10e9;
```

- Plot the active element pattern as a function of θ .
- At what angle of incidence θ_{in} is a higher order Floquet mode entering the visible region, as predicted by theory?
- Plot the total patterns for normal incidence $\theta = 0^\circ$ and for oblique incidence $\theta = \theta_{\text{in}}$
- How high is the level of the grating lobe for oblique incidence $\theta = \theta_{\text{in}}$ compared to the main lobe and why?